

Artificial Intelligence

Short Questions

Q1. Visual recognition is inherently challenging for a computer. Enlist at least three challenges

1. Variations in Appearance

- Objects can look very different depending on the lighting, angle, or background.
- For example, a cat in the sunlight looks different from a cat in the shadow—or even when it's curled up vs. standing.
- Computers struggle to recognize it's the *same* cat every time.

2. Hidden Objects

- Sometimes, part of an object is blocked or covered by something else.
- Imagine trying to recognize a bicycle in a photo when half of it is behind a wall.
- A human can guess what's behind and still understand—it's harder for a computer.

3. Intra-Class Differences

- The same type of object can appear in many forms.
- Think of all the different shapes and colors a chair can have—wooden, plastic, with arms, without arms.
- Computers need to learn that all of these still count as “chair.”

Q2. Explain the concept of backpropagation in context of training an ANN

What is Backpropagation?

Backpropagation (short for *backward propagation of errors*) is the learning method used by neural networks to improve themselves. It's like how we learn from mistakes—by figuring out what went wrong and adjusting next time.

How It Works (Step-by-Step):

1. Forward Pass

- The input data (like an image or number) passes through the network.
- Each layer processes it and gives an output (called prediction).

2. Calculate Error

- We compare the network's output with the correct answer (also called the label).
- The difference is called the **error** or **loss**.

3. Backward Pass (Backpropagation)

- Now the real magic happens! The network works **backwards** from the output to the input, figuring out **which connections (weights) caused the error**.
- It uses a bit of math (called derivatives) to figure out how much each weight contributed to the mistake.

4. Update Weights

- The network **updates the weights** (the strength of connections) using a small correction. This helps the network do better next time.

And Then We Repeat...

This process happens again and again—thousands of times—until the network becomes really good at giving the right answers.

A Simple Analogy Example:

Imagine trying to shoot an arrow at a target with your eyes closed.

- You shoot (forward pass), then open your eyes and see you missed.
 - You measure how far off you were (error).
 - You think about what you did wrong (backpropagation).
 - Next time, you adjust your aim (update weights).
- After enough tries, you start hitting the bullseye!

Q3. A city is trying to optimize traffic lights timings to reduce congestion during peak hours. How would you apply a genetic algorithm to optimize the timing of traffic lights across multiple intersections? What factors would influence your fitness function, and how would you address real world constraints like pedestrian crossings?

The Problem

During rush hours, there's a lot of traffic and the city wants to reduce waiting time at red lights. The idea is to find the best timing for traffic lights so cars move smoothly through intersections.

What's a Genetic Algorithm (GA)?

Think of it like a game where each “player” is a traffic light plan. We try different plans, see which ones work best, and use them to make even better ones—just like improving a recipe by tasting and tweaking!

Step-by-Step Explanation

1. Create Traffic Light Timings (Like Making Candidates)

We make different versions of traffic light timings for each intersection—for example:

- Green: 30 sec, Yellow: 5 sec, Red: 25 sec
- Or another one: Green: 45 sec, Yellow: 4 sec, Red: 31 sec

Each full plan is like a "solution" or “candidate.”

2. Test Each Plan (Using Fitness Function)

We simulate how traffic would move with each plan. The best ones:

- Reduce car waiting times
- Avoid traffic jams
- Don't make people wait too long to cross the road
- Help traffic flow better at nearby roads

3. Keep the Best, Mix & Improve

We:

- **Pick the best plans** that worked well
- **Mix parts** of two good plans together (like combining ideas)
- **Make small changes** randomly (called mutation) to try something new

This gives us new plans that might work even better!

4. Repeat the Process

Keep testing, mixing, and improving until we find the best traffic light schedule.

How We Handle Real-Life Stuff

- **Pedestrian Crossings:** We make sure people always get a chance to cross safely.
- **Min/Max Times:** We set limits so lights aren't too short or too long.
- **Connected Roads:** We make sure nearby signals work together to avoid constant stopping.

Q4. What are the key components of First Order logic and how are they used to represent knowledge ?

Key Components of First-Order Logic (FOL)

1. Constants

- These are specific objects or names.
- Example: Ali, Car1, Pakistan
- Think of them as "fixed names" for things.

2. Variables

- These are placeholders for any object.
- Example: x , y , z
- Like in math, they can stand for anything.

3. Predicates

- These describe properties or relationships between objects.
- Example:
 - $\text{Tall}(\text{Ali})$ means "Ali is tall"
 - $\text{Loves}(x, y)$ means "x loves y"
- They're like verbs or adjectives that give meaning.

4. Functions

- Functions give us an output based on input(s).
- Example: $\text{FatherOf}(\text{Ali})$ might return Ali's father.
- They help break down complex ideas.

5. Quantifiers

- They tell us about the quantity of objects we're talking about.

Symbol	Name	Meaning
$\forall$	Universal	"For all" (e.g. all people)
$\exists$	Existential	"There exists" (e.g. some cats)

6. Logical Connectives

- These join ideas together, like in logic puzzles.

Symbol	Name	Example	Meaning
$\wedge$	AND	$\text{Tall}(x) \wedge \text{Smart}(x)$	x is tall and smart
$\vee$	OR	$\text{Happy}(x) \vee \text{Sad}(x)$	x is happy or sad
$\neg$	NOT	$\neg \text{Rich}(x)$	x is not rich
$\rightarrow$	Implies	$\text{Student}(x) \rightarrow \text{Studies}(x)$	If x is a student, then x studies
$\leftrightarrow$	Biconditional	$A \leftrightarrow B$	A is true if and only if B is true

How FOL Represents Knowledge

- We use **FOL statements to describe real-world facts** in a formal, logical way.
- These statements let a computer **infer new facts** based on existing ones.

Example:

Let's say we have:

1. $\forall x (\text{Cat}(x) \rightarrow \text{Animal}(x))$ – All cats are animals
2. $\text{Cat}(\text{Whiskers})$ – Whiskers is a cat
 - From this, the system can **logically deduce**: $\text{Animal}(\text{Whiskers})$

This is how AI systems use logic to “think” step by step.

Q5.What are the limitations of Fuzzy logic and how they can be addressed in complex systems?

While **Fuzzy Logic** is powerful for handling uncertainty and mimicking human reasoning, it does have some limitations—especially in large or complex systems.

Limitations of Fuzzy Logic & How to Address Them

1. Lack of Learning Ability

- **Problem:** Fuzzy logic systems don’t learn on their own. You have to define all the rules and membership functions manually.
- **Solution:** Combine fuzzy logic with learning systems like **Neural Networks** or **Genetic Algorithms**. This hybrid approach allows the system to adjust rules over time based on data.

2. Rule Explosion (Too Many Rules)

- **Problem:** As the number of input variables increases, the number of possible rules grows quickly—making the system complex and hard to manage.
- **Solution:**
 - Use **rule pruning** to eliminate unnecessary or redundant rules.
 - Apply **modular fuzzy systems**, where smaller fuzzy subsystems handle different parts of the problem.

3. Difficult to Design Membership Functions

- **Problem:** It’s not always easy to define “how fuzzy” something is. Getting the shape of membership functions right (like how steep or wide) can be tricky and affect the system’s performance.
- **Solution:**
 - Use expert knowledge where available.
 - Apply optimization techniques (like **Particle Swarm Optimization** or **Genetic Algorithms**) to tune them automatically.

4. Not Always Precise or Scalable

- **Problem:** Fuzzy logic gives approximate results, which may not be suitable when **high precision or scalability** is needed.
- **Solution:**

- Use fuzzy logic in the **front-end** for decision-making, and let **crisp, numerical methods** handle the back-end computations.
- Or combine it with **probabilistic methods** when precise uncertainty modeling is needed.

5. Hard to Validate and Test

- **Problem:** Since fuzzy systems are based on vague concepts, it's difficult to verify if they behave correctly under all conditions.
- **Solution:** Use **simulation tools**, real-time testing, and **sensitivity analysis** to fine-tune and validate the system before applying it in the real world.

Q6. How is NLP and AI related? Discuss different applications of NLP in detail.

How is NLP Related to AI?

Natural Language Processing (NLP) is a *branch of AI* that helps computers understand, interpret, and respond to human language.

Think of **AI** as the big umbrella and **NLP** as one of its spokes focused on **language**. If AI is about making machines smart, then NLP is about making them smart with **words, sentences, and meaning**.

Applications of NLP (With Easy Examples)

1. Chatbots and Virtual Assistants

- Tools like **Siri, Alexa**, or even **Copilot** use NLP to understand your questions and answer naturally.
- **Example:** "What's the weather today?" → Your assistant understands and replies with a full sentence.

2. Text Summarization

- NLP helps shorten long documents into quick summaries.
- **Used in:** News apps, research tools, or education platforms.
- **Example:** Turning a long article into 3 key bullet points.

3. Spam Email Detection

- NLP scans the content of emails to detect spam based on keywords and patterns.
- **Example:** Emails with "You've won a prize!!!" are likely spam.

4. Language Translation

- Tools like **Google Translate** use NLP to convert sentences from one language to another.
- **Example:** "How are you?" → "آپ کیسے ہیں؟"
- It's more than word-for-word—it tries to understand the *meaning* too.

5. Speech Recognition

- Converts spoken words into written text using both **NLP and speech processing**.
- **Used in:** Voice typing, meeting transcription, language learning apps.
- **Example:** You speak “Hello,” and it types “Hello” on your screen.

6. Sentiment Analysis

- NLP can tell if someone is feeling **positive**, **negative**, or **neutral** in their message.
- **Used in:** Customer reviews, social media monitoring.
- **Example:** “I love this product!” → Positive
“This was terrible!” → Negative

7. Search Engines

- When you type something into a search bar, NLP helps the system figure out what you *mean*, not just what you typed.
- **Example:** “Best pizza near me” → It knows you're hungry and nearby, not just looking for pizza facts!